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BCS-4C

Short Questions

Q1:

The three advantages of database system over file-based system are:

1- **Data Redundancy:** means storing of same data multiple times in different files which leads to multiple effort and alot of space is wasted in file-base system which is not the case in DBMS

2- **Reduce data inconsistency:** due to data redundancy in file ~~m~~ based system, files that represent same data may be in-consistent and hard to manage as compared to DBMS.

3- **Providing Backup and Recovery:** A DBMS has the facility of recovering fom hardware or ~~backup~~ software failure and it provides backup and recovers and restores the database if the system fails in between the transcation.

Q2:

The primary role of database management system is to efficiently and securely store, retrieve and update ~~data~~ informatiom.

Data Manipulation i.e; insert, update, delete is done through query language. The access and process is done by sending queries and requests.

The DBMS also has to give protection to database from hardware or software failures ~~of~~ and unauthorized access.

Q3:

Data model is collection of concepts to describe structure of database. It provides abstraction and constraints of the database. It is visualization before implementation

It has three concepts; Entity, Attributes and Relationship.

- Entity is the realworld concept that we will implement in database, i.e. student
- Attributes represents properties of entity, i.e, student name.
- Relationship between two or more entities such as there is registration relationship between student and courses.

Q4:

It can be defined as capacity to change the schema at only one level that it do not effect the program and we don't have to change the schema at higher level.

It provides abstraction and hides the details, which allows programmers to modify the definitions without affecting the program that use it.

There are two types of data independendence

1- Logical data independence: to expand, change constraints or reduce database without changing external schema.

2- Physical data independence: modifying physical scheme without changing conceptual schema.

Q5: - Database Administrators (DBA): The DBA are those who control and manages the database. They are responsible for authorizing access to database, monitoring the use of database, they also manage hardware and software resources and also maintaing the integrety and privacy of database and protect from security breaching.

- End Uses: are the people who will access the database. They may need the database for updating or generating reports, The database is created for these users. End users can be engineers who need DBMS to implement their applications, stand alone user maintain personal databases. casual users have low level of interaction and uses application to access information

Q6:
- DDL: It is data definition language. It helps to define the structure of the database. It's set of ~~SQL~~ command that create, modify and delete the database structure. Some basic commands are;
  - CREATE : used to create a new database, table
  - ALTER : used to modify tables, columns, datatypes and constraints
  - DROP : used to delete column, table, database or constraints

- DML: It is data manipulation language to manipulate data in database. It's commands are:
  - SELECT: used to read/retrieve data from one or more tables
  - INSERT: used to add new record in table
  - UPDATE: used to modify existing record from table
  - DELETE : used to remove/delete record from table

Q7:

Data encapsulation is a concept of object oriented. Programming it is a way to bindle the implementation ~~details from the user~~ of methods and data (attributes and properties within a single unit. The unit is called object in OOP and entity in DBMS.

Tables can have different fields with different data types which are attributes of that object and a record is ~~as~~ single object.

Objects can be stored in database using object relational database.

Q8:

Data: known facts and figures that are raw is called data

Example: 22F3277 is a data

Information: processed form of data ~~with~~ that is meaningful and has context is called information.

i.e. 22F-3277 is roll no of a student at FAST.

Knowledge: It is the understanding of the information.

i.e. It is roll no of student of 22 batch of Faisalabad campus is knowledge

Meta data: It is data about data. It defines actual data. The ~~med~~ metadata is a catalog and describe structure of data. It defines the (constraints, datatypes, lengths, minimum value, should field be left empty or not etc.). of field of the data base.

Q9:

Self describing data refers to type of data that includes meta data about it's format or structure. The metadata allows the data to be interpreted.
The self describing aspect allows to interpret and process the data with additional documentation.

Q10:

For this example 3 tier structure would be more preferable because it provide scalability and can be managed easily (changes to user interface can be made independently) and also it enhances security and provides flexibility.

LONG QUESTION

Q1:

The main perrpose of 3 schema archtecture is to separate use application from physical database.

1. **External Schema:** It represents the user's view of data. It defines how data is presented and accessed by each individual user. Each application may have different external schema. This allows user to interact with DBMS without understanding of complex level.

2- Conceptual Schema ~~(Logical~~: It describes the structure of whole database for community users.

It hides the details of physical storage structure. It main purpose ~~its~~ to describes entities, data ~~base~~ types, relationships, It also describes use operations and constraints. It describes data model.

3- Internal level Schema

→ it is low level structure of the Architecture.

→ It represents the physical storage structure and organization. It give complete details of the access paths.

→ To improve performance and storage efficiency changes are made here.

Data Abstraction Support:

→ The three levels allows data abstraction by hiding details which are not neccessary to be shown

→ It also hide details within different levels.

→ Changes in one level doesnot effect the other levels. it support independence b/w logical and physical aspect of database.

Q2:

2-tier:

In 2-tier their are two components, client and the server. The client directly communicates with server. it is easier to build. It runs slower.

i) **File sharing System:** In basic file sharing system clients directly interacts with file server to perform basic operation like ~~feet~~ read, write, share. Server handles storage

ii) **App with local database:** desktop application with local database where client interact directly with database.

iii) **Reservation System:** for small business uses 2-tier where client ~~di~~ application directly interact with local database to make reservation.

3-tier:

In 3-tier there are thee components: client, applicationserver and database server. It is common for web application It provides additional features as security, encryption, decryption.

→ **Online Banking System:** user interact with app to access banking service. The application server manages process the transcation, and database server mange financial record data.

→ **Online Shopping System:** clients browse through application the application server manages user authentication, order processing and database server holds product details and user profile

iii) **web-application:** User interact with web platform, the ~~authentication~~ application server manages process and access server.

N-tier Architecture:

It extends 3-tier by introducing additional layers for improving scalability, flexibility and to maintain easily.

i) **Healt care Information System:** It can have extra layer such as laboratories or insurance provider.

ii) **Online Learning Mangement System:** additional layers can be assesments, or video streaming.

iii) **Online Shopping (E-commerce):** A shoping platform with additional ~~fe~~ layer as cart, product details. are implemented.

Q3: i) Command Line Interface:

It is text based based where user interacts with DBMS by typing commands. User can execute queries and manage data structures.

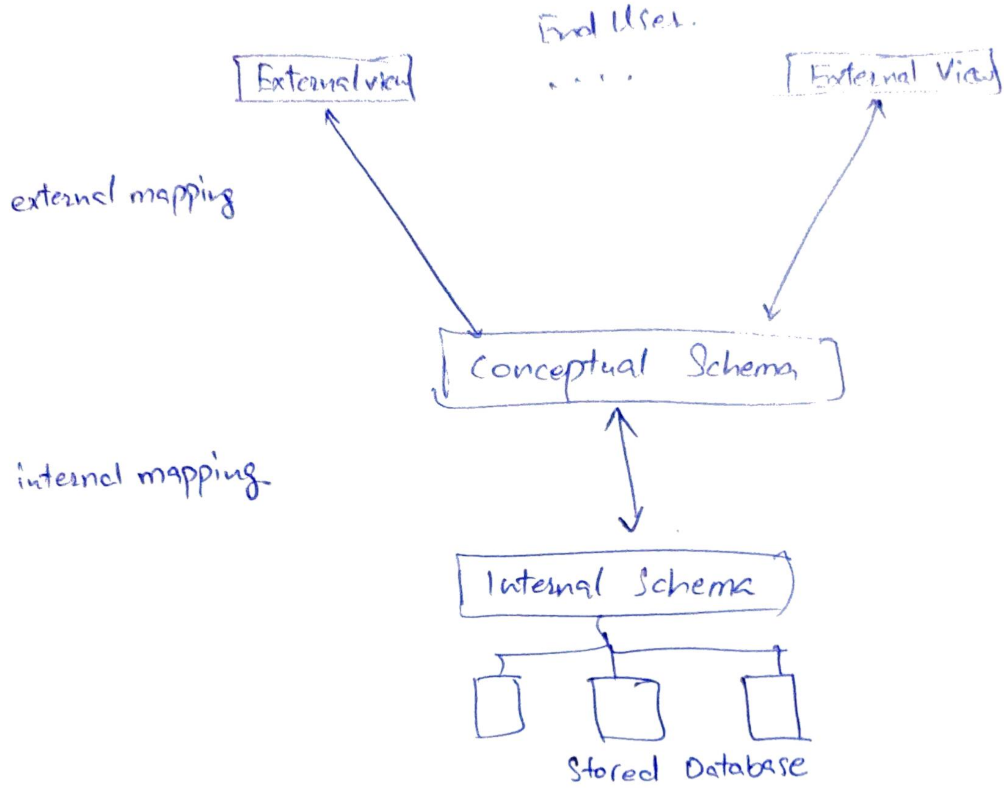
ii) Graphical User interface: It provide a visual interface and users can interact with buttons, menu and visual appealing ~~eleme~~ design making more use friendly.

iii) Web-based interface: Some DBMS offer web base access through web browser so that it can be accessed anwhere remotedly. and are accessible from any device. The interface is user-friendly.

(iv) Application Programming Interfaces: An API provide set of functions that developers can use to integrate ~~to~~ database functionalities into their applications.
It is common is web applications, mobile apps.

(v) Query Language Interface: It is standard language for Interacting with relational database. ~~operatio~~ Users developer, can use SQL commands to performs operation such as ~~quer~~ updating, and managing the database

Q4:



The internal schema is low level and is directly connected with database. It represent physical ~~struc~~ storage structure and gives the details of access paths. any chances made here are done to improve performance and storage efficiency. In conceptual schema it defines the ~~structu~~ entities, datatypes and relationships and constrants. and hide internal schema details. The upper level is external view it is different of different users. It defines how data is accessed by each individual.
The DBMS transfer request from external schema to conceptual schema and from there to internal schema and the relevant data according to ~~the~~ user is then send back. this process of transforming request and result between levels

are called mapping. Change in any level does not effect to change other level. It provides clear separation.

Q5: Data Independence.

It can be defined as the capacity to change the schema at only one level that it does not need to change the schema at next higher level. It provides abstraction and hide the details which allows the programmers to modify the definition without affecting the program.

There are two types of data independence:

1- Logical data independence: allows for changing conceptual schema without having to change external schema. We ~~can~~ can change conceptual schema to expand, change constraints or reduce database.

2- Physical data independence: it enables changes to physical schema without changing conceptual schema. It defines how the data is physically stored. and it access paths., changes made here can be done without impacting conceptual schema, by changing file structure, to improve the performance of retrieval or update.